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Behind Schedule - Capturing the Effects and Side Effects of an Intervention on the Learning Process

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Abstract. Educational interventions often produce effects beyond their intended outcomes, yet these side effects remain understudied. This controlled experiment examined how late-of-schedule notifications affected learning behaviors in an online project management course with 105 Finnish business students. Students falling behind schedule were randomly assigned to receive reminder notifications (n=57) or standard communication (n=48). Using sequence analysis, we identified behavioral patterns distinguishing the groups. The experimental group demonstrated markedly increased self-regulatory behaviors, particularly feedback-seeking and self-assessment patterns. These findings reveal that simple scheduling reminders can trigger profound metacognitive shifts, transforming surface-level task completion into deeper, reflective learning engagement.

Keywords: learning analytics; self-regulated learning; sequence analysis; intervention; time management.

1 Introduction

Higher education, especially in online learning settings, requires well-developed self-regulation skills, enabling students to take control of their learning process and achieve academic success. Moreover, the quality of students' self-regulation is causally related to many other important outcomes, such as well-being, motivation, and performance, also in their future career [1].

In this study, we focus on a subset of the self-regulated learning (SRL) skills, namely those related to planning and monitoring the learning process, i.e., setting a schedule and adhering to that schedule, aspects of SRL that students often struggle with [2]. A variety of interventions have been studied to support students in demonstrating these skills and helping them be 'on time' [3, 4], each having hypotheses on the effects on the (self-regulated) learning process and outcomes. While previous studies have typically focused on primary outcomes without

systematically examining potential side-effects or unintended consequences on broader learning processes. Furthermore, few studies leveraged learning analytics (LA) to provide detailed behavioral insights into how students actually respond to these interventions beyond simple completion metrics. This gap limits our ability to design more effective interventions and understand their full impact on students' learning. We posit, however, that even the simplest of interventions will have effects beyond those encompassed in these hypotheses. Such unintended, and possibly indirect, side effects are often overlooked in evaluation, but could have important implications for students involved.

Hence, in this study, we conducted a controlled experiment in an online course, where students in the experiment group received a notification email to remind them of their self-set deadline, in case they missed it. Using LA techniques, we dig deep into the resulting learning processes, comparing the intervention and control groups to identify both effects and side effects. With the results of these analyses, we aim to answer our main research question: What are the effects and side effects of late-of-schedule intervention for learning behavior in an online learning environment? Moreover, we aim to demonstrate the affordances of LA methods for a more comprehensive evaluation of (technology-enhanced) interventions targeting SRL by students in higher education.

2 Background

2.1 Self-regulated learning in online environments

SRL is vital for academic success in online learning environments, as learners must independently manage their cognitive, metacognitive, and motivational resources in the absence of traditional classroom structures. Effective SRL encompasses goal-setting, time management, self-monitoring, and the adaptive use of feedback. Online settings uniquely challenge learners due to their reliance on asynchronous activities, requiring higher levels of autonomy and proactive behavior. Empirical research highlights that conscientiousness and general academic competence significantly influence the effective application of SRL strategies, especially when external structures like deadlines alone do not prevent behaviors such as cramming [5]. Process mining studies have identified successful time management strategies, including thorough preparation, periodic revisiting, and adaptive regulation activities, as critical factors influencing academic outcomes [3]. While targeted LA interventions show potential for scaffolding SRL processes and improving outcomes, their effectiveness varies considerably [6], underscoring the importance of aligning interventions with theoretical frameworks such as Zimmerman's cyclical model of SRL [7]. Recent findings suggest that mobile-based push notifications can significantly enhance students' engagement and self-regulatory behaviors, effectively reducing academic procrastination and supporting autonomy in digital learning contexts [4]. However, a systematic review indicates that less than half of LA interventions report positive outcomes, emphasizing the need for deeper exploration of how specific analytics-driven strategies can effectively support all phases of the SRL process [8]. Thus, advancing our understanding of these targeted

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interventions is essential for enhancing SRL support in online learning environments.

2.2 Learning analytics interventions

LA refers to "the measurement, collection, analysis, and reporting of data about learners and their contexts, for purposes of understanding and optimizing learning and the environments in which it occurs" [9]. Early learning management systems (LMS) applications counted online activities to identify at-risk students. As research has matured, emphasis has been placed on explaining why and how to support students to "diagnose" what happened and offering them a prescription for how to proceed to succeed. Recently, the focus has shifted from counts and predictive models to processes and mechanisms, as counts reveal little about how events unfold or progress. Charting the learning process has gained momentum as it shows how students work, when they struggle, and what leads to improvement. Understanding this process helps explain how interventions work or why they fail.

Several methodological approaches have been employed to capture the learning process, most of which capture time, order, or relationships. For instance, sequence analysis has been extensively used to capture the order of learning activities students perform while learning [10]. Sequence analysis is usually complemented with other statistical techniques to capture the patterns within the sequences that are indicative of a certain tactic or strategy. For instance, using sequence analysis, researchers found patterns of how students self-regulate their learning, how they organize their time, or manage their online assignments [10].

LA interventions aim to support students' SRL in online settings by leveraging their digital trace data for feedback and guidance [11]. These interventions often provide imbalanced support for the cyclical phases of SRL [cf. 12]. To realize their potential, LA interventions must be integrated with SRL theory to effectively foster the necessary metacognitive and behavioral skills (e.g., time management) for online success. Moreover, the evaluation of such interventions should adopt a more holistic view [13], in particular by not only looking for direct, intended effects but also capturing indirect, unintended side effects.

3 Methods

This study employed a controlled trial design to examine an intervention aimed at supporting learning regulation. The participants were drawn from a cohort of Finnish bachelor-level business students enrolled in an online project management course. At the start of the course, all students were randomly assigned to either an experimental or a control group using the randomization feature of the LMS. The intervention targeted students who subsequently fell behind their planned study schedule. This resulted in a final study sample of 105 students, who retained their original group assignments from the initial randomization (experimental group,

n=57; control group, n=48). At the beginning of the course, students in the experimental group received targeted feedback designed to support their learning regulation. Informed consent was obtained from all participants before the study.

Student time-stamped learning activities (n=40,650) captured through the Moodle LMS were recoded to simplify the analysis of the granular trace data. The recoded data [14, 15] compressed the data into fewer categories: visits to the course page, learning resource views, assignment viewing, checking assignment criteria, assignment submitting, performing self-evaluation, and viewing feedback (see Table 1).

Students were prompted to set individual deadlines for 11 assignments throughout the course. These individual deadlines and the corresponding actual submission timestamps were systematically recorded. Comparison of deadlines and submission timestamps enabled the classification of submissions as on-time or late, providing measures of schedule adherence.

Table 1. Distribution of the activities before sequencing.

Activity	n	percent
Assessment criteria (Crit)	1554	0.038
Assignment viewed (AV)	8720	0.215
Course page	7355	0.181
Evaluating	1338	0.033
Feedback	1590	0.039
Learning Resource viewed (LR)	5060	0.124
Self-evaluation (SE)	4676	0.115
Submitting	10357	0.255
Total	40650	

The intervention for the experimental group was administered manually by the course teacher, who monitored student progress against the personal schedules they had set for the 11 course assignments. When a student was observed to be falling behind their self-determined timeline, the teacher sent a personalized email. This communication noted the delay and offered support by asking if the student needed help with their studies or assistance in revising their course plan [16]. In contrast, control group students with similar late submission patterns received only regular course communications (e.g., general course announcements, assignment instructions, and administrative updates) without any personalized deadline reminders. For analytical purposes, control group students were assigned virtual intervention timestamps matching when they would have received notifications if in the experimental group, enabling direct behavioral comparisons between conditions.

Raw activity traces were processed using R to create meaningful learning sessions using temporal thresholds. Learning sessions were defined using a 30-minute

temporal gap criterion [17]. The data collection period spanned the entire course duration, resulting in 4,039 individual learning sessions (experimental group: 2,272 sessions; control group: 1,767 sessions) analyzed using the TraMiner package [18]. The median session length was 24 actions, with experimental group sessions averaging 27 actions and control group sessions averaging 22 actions. Sessions were further filtered to exclude outliers, retaining only sessions with 2-100 actions to focus on substantive learning episodes.

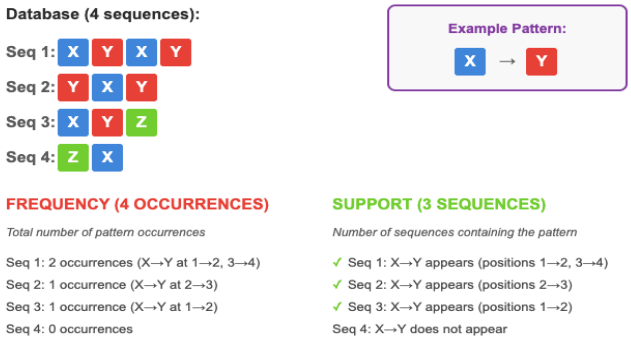


Fig 1. Illustration of sequence patterns with frequency and support metrics. A database containing four sequences (Seq 1-4) with elements X, Y, and Z is analyzed for the occurrence of pattern X → Y. **Frequency** measures the total number (including multiple appearances in the same sequence) of pattern occurrences across all sequences (4 occurrences), while **support** measures the number of sequences containing at least one occurrence (no double counting within a single sequence) of the pattern (3 out of 4 sequences). The example demonstrates how the same pattern can appear multiple times within a single sequence (e.g., twice in Seq 1) and how some sequences may not contain the target pattern at all (Seq 4).

The differences in learning pattern subsequences (i.e., ordered events or actions that unfold sequentially forming a pattern) between experimental and control groups after the intervention were analyzed using the tna package [19]. Both frequency and support statistics were computed to assess the total number of pattern occurrences and the number of sequences containing these patterns across the two groups (Fig. 1). To statistically identify subsequences that distinguish the experimental and control groups, chi-squared tests with Bonferroni correction were applied. A maximum pattern length of 5 actions was used for the analysis. This approach ensured that significant subsequences were identified and tested for statistical significance, with the Bonferroni correction mitigating the risk of false positives. We classified these patterns as either passive or active [20].

4 Results

The overall sequence distribution plot (Fig. 2, left) represents the development of learning sessions over time. Different colors represent different activities learners perform while studying online. The distribution of the activities is quite well balanced. The course page is more commonly visited at the beginning of the session, while the proportion of submissions increases as the sessions unfold. When subsequences (Fig. 2, right) are analysed, we can see significant differences. For example, Feedback-SE patterns are found more frequently than expected from the sequences of the experimental group students.

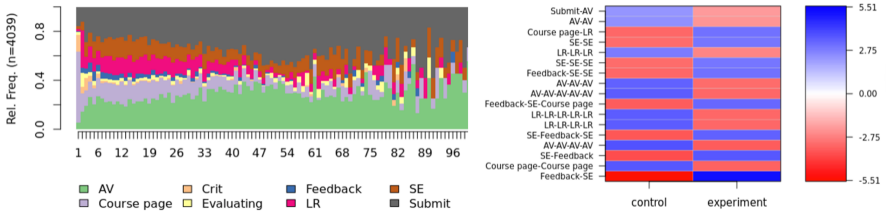


Fig 2. Left: Sequence distribution plot of the course sessions (n=4039). The X-axis represents the order of the steps (0-100) during sessions, and the Y-axis represents the relative proportion of actions during a particular step of the sequence. **Right**: Discriminating subsequences between the control and experiment groups. Blue indicates patterns that occur more frequently in a group than would be expected if there were no group differences, while red indicates patterns that occur less frequently than this baseline expectation.

Statistical analysis (Table 2) revealed seventeen patterns with significant differences between groups after Bonferroni correction ($\chi^2 = 11.51$ to 53.72, all $p < 0.05$). All active patterns are more prevalent in the experimental group. The most pronounced differences emerged in (active) self-assessment and feedback behaviors. The experimental group exhibited markedly higher engagement with feedback-seeking activities, particularly the Feedback-SE pattern, which occurred 6.75 times more frequently in the experimental group than in controls (108 vs 16 occurrences, $\chi^2 = 53.72$, $p < 0.001$). Similarly, the SE-Feedback pattern showed a ninefold increase in the experimental group (45 vs 5 occurrences, $\chi^2 = 24.77$, $p < 0.001$). These patterns suggest the experimental group developed more active feedback-seeking behaviors and integrated self-assessment more systematically into their learning process.

Conversely, the control group demonstrated higher frequencies of repetitive instructional behaviors. The four-fold repetition of instruction patterns (AV^4) occurred exclusively in the control group (22 vs 0 occurrences, $\chi^2 = 23.18$, $p = 0.002$), as did the five-fold repetition pattern (20 vs 0 occurrences, $\chi^2 = 20.74$, $p = 0.014$). This suggests control group learners engaged in more intensive instruction review, potentially indicating difficulty with initial comprehension, or they don't feel the urgency of being late.

The experimental group showed substantially higher engagement with basic self-assessment activities, with both SE^2 and SE^3 patterns occurring more frequently (872 vs 1201 for SE^2, $\chi^2 = 13.93$, $p = 0.012$; 310 vs 481 for SE^3, $\chi^2 = 16.39$, $p = 0.018$).

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Table 2. Significant Differences in Learning Pattern Frequencies Between Control (Freq_C) and Experimental (Freq_E) Groups.

Pattern	Activeness	Freq_C	Freq_E	χ^2	P_adj
Feedback-SE	Active	16	108	53.72	<0.001***
Course page-Course page	Passive	784	711	25.3	<0.001***
SE-Feedback	Active	5	45	24.77	<0.001***
SE-Feedback-SE	Active	0	28	22.38	<0.001***
AV^4	Passive	22	0	23.18	0.002**
Feedback-SE-Course page	Active	2	32	20.81	0.002**
AV^3	Passive	28	4	20.19	0.002**
LR^4	Passive	131	77	22.08	0.003**
Feedback-SE-SE	Active	0	22	17.2	0.012*
SE^2	Active	872	1201	13.93	0.012*
LR^5	Passive	85	41	21.07	0.012*
AV^5	Passive	20	0	20.74	0.014*
SE^3	Active	310	481	16.39	0.018*
Course page-LR	Passive	1375	1820	12.73	0.023*
AV^2	Passive	332	295	11.92	0.036*
LR^3	Passive	218	170	14.9	0.039*
Submit-AV	Active	2879	3111	11.51	0.044*

Note: All comparisons used Chi-squared tests. p_adj = Bonferroni adjusted p-values. Superscripts indicate pattern repetitions. *p<0.05, **p<0.01, ***p<0.001

Learning resource engagement patterns revealed interesting group differences in depth of interaction. While the control group showed higher frequencies of extended learning resource sequences (LR^4 and LR^5 patterns: 131 vs 77 and 85 vs 41, respectively), the experimental group engaged more broadly with diverse learning activities. This suggests the control group pursued deeper, more concentrated engagement with specific resources, while the experimental group adopted a more distributed approach across multiple learning modalities.

Table 3 presents a support-based analysis to examine the prevalence of patterns across learning sessions. While Table 2's frequency analysis counted total pattern occurrences, support measures the proportion of sessions containing each pattern at least once (see Fig. 1), revealing whether behaviors reflect intensive repetition by a few students or widespread adoption across groups. Positive support differences

indicate patterns more prevalent in the control group; negative values show experimental group dominance.

The control group shows notably higher numbers of sessions where they engaged with navigation patterns like "Course page-Course page" (+0.058) and feedback-seeking behaviors like "Submit-Feedback" (+0.052). Conversely, the experimental group demonstrates higher numbers of sessions with patterns in direct learning resource engagement, particularly "Course page-LR" (-0.050). Most notably, “Feedback-SelfAssess” was four times more prevalent in the experimental group (-0.031), aligning with the frequency analysis findings of enhanced self-regulatory behaviors.

Table 3. Subsequences with the highest differences in occurrences between groups. The *pattern* column shows sequences of learning activities, The *support* columns (C and E) indicate how frequently each pattern occurs within each group, expressed as proportions of sessions, *support diff* is the absolute difference in pattern frequency, *support ratio* is Support C’s frequency to Support E’s frequency, and *relative support* is the percentage change in pattern frequency relative to Group E’s baseline.

Pattern	Support C	Support E	Support diff	Support ratio	Relative support
Course page-Course page	0.220	0.162	0.058	1.353	0.151
Submit-Feedback	0.243	0.191	0.052	1.270	0.119
Course page-LR	**0.406**	**0.456**	**-0.050**	**0.890**	**-0.058**
Submit-AV	0.467	0.419	0.047	1.113	0.054
Submit-Feedback-AV	0.200	0.156	0.045	1.284	0.125
Feedback-AV	0.203	0.161	0.042	1.256	0.114
Submit-AV-Submit	0.242	0.202	0.040	1.198	0.090
AV-Submit	0.388	0.350	0.038	1.108	0.051
Submit-AV-Submit-AV	0.194	0.156	0.038	1.241	0.108
AV-Submit-AV	0.226	0.191	0.035	1.181	0.083
AV-Course page	0.181	0.149	0.032	1.212	0.096
Feedback-SelfAssess	**0.009**	**0.040**	**-0.031**	**0.245**	**-0.631**
Course page-Course page-Submit	0.069	0.039	0.030	1.744	0.276
Course page-Submit	0.175	0.146	0.029	1.199	0.091
Evaluating-Criteria	0.121	0.092	0.029	1.307	0.134
Submit-AV-Course page	0.138	0.112	0.026	1.233	0.105
Submit-Submit	0.193	0.169	0.024	1.144	0.068
Submit-Submit-AV	0.173	0.149	0.023	1.156	0.073
AV-Submit-Submit	0.184	0.161	0.023	1.144	0.068
AV-Submit-AV-Submit	0.166	0.144	0.022	1.151	0.071

The intervention group engaged in more repetitive assessment and feedback behaviors, likely driven by the intervention itself, as seen in the higher support and

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frequency of self-assessment and feedback patterns. This suggests the intervention actively prompted more cycles of reflection and feedback. In contrast, the control group was more focused on learning resources, with higher support for LR patterns and more consistent use of assessment activities across sessions. The control group showed less repetition, indicating a more straightforward approach to learning. The intervention group's behavior was shaped by the intervention's design, promoting deeper reflection through feedback repetition.

5 Discussion

The experimental group's increased engagement with self-regulatory behaviors—particularly the 6.75-fold increase in Feedback-SE patterns and ninefold increase in SE-Feedback patterns—indicates that late-of-schedule notifications triggered metacognitive strategies. Students receiving notifications actively sought feedback and systematically integrated self-assessment into their learning processes. This shift aligns with the models of SRL, where external cues can act as scaffolds that prompt learners to move from performance oriented actions to metacognitive monitoring and reflection [12, 21]. The unique contribution of sequence analysis here is its ability to reveal not just an increase in discrete self-assessment actions, but the formation of these meaningful, structured behavioral chains (Feedback-SE), providing tangible evidence of a deeper cognitive process.

Conversely, the control group's engagement with repetitive instruction viewing (AV^4, AV^5) and extended resource consultation (LR^4, LR^5) suggests strategies focused on diverse activities and learning resources. This behavior may indicate engagement with the task with less emphasis on the assessment and is characteristic of students who, in the absence of targeted support, may resort to less effective strategies like simple rereading [20]. These learners adopted a more content-focused approach, investing more time in reviewing materials but potentially without the metacognitive framework that was triggered in the experimental group's behavior. Pattern detection capability of our method allowed us to precisely identify and quantify these repetitive loops (e.g., AV^5), offering concrete evidence of less productive strategies that simple activity counts would miss.

The mechanism by which a simple scheduling reminder produces such behavioral changes requires careful consideration. We propose interrelated explanations for why being notified of lateness would specifically enhance self-evaluation and feedback-seeking activities. First, the intervention may have functioned as a metacognitive awakening - by explicitly highlighting the gap between planned and actual progress, the notification prompted students to evaluate their learning process rather than continuing on autopilot, a core component of self-reflection phase in the cyclical SRL models [12]. This awareness likely prompted students to question not just when they were completing tasks, but how well they understood the material, leading to increased self-evaluation activities. Bear in mind, that students were prompted to set individual deadlines rather than obeying deadlines that were set

externally—this might be a prerequisite for such an effect to occur [4]. Second, the notification may have triggered a strategic recalibration of learning approaches. But that also came at the expense of more repetitive assessment-related interactions and less focus on learning resources. As such, intervention may also have some side effects that can be summarized as an emphasis on assessment at the expense of learning and repetitive behavior at the expense of confidence or efficacy.

Meanwhile, the control group's sustained focus on learning materials—while different—represents an alternative approach rather than a purely passive strategy. These findings illustrate how interventions create complex side effects: the scheduling prompts not only addressed time management (the intended effect) but also altered how students allocated their learning efforts, sometimes at the expense of content engagement. This supports the theoretical position that self-regulation benefits from external scaffolding [21], while highlighting that such scaffolding can produce unexpected redistributions of learning activities in online environments.

Several limitations should be considered when interpreting these findings, which also point toward avenues for future research. First, this study was conducted within a single project management course, and the findings may not be generalizable to other academic disciplines or learning contexts. Replicating this experiment in diverse courses and with larger student cohorts is necessary to validate the observed effects. Second, our analysis is exclusively on log-file data from the online learning environment, which does not capture offline study behaviors like reading physical textbooks or peer discussions. Future work could integrate qualitative methods or self-report measures to gain a more holistic view of student learning strategies.

Furthermore, the repeated questionnaires used in this study may have inadvertently acted as a co-intervention. The act of regularly prompting students to reflect on their learning could have influenced their metacognitive processes, a phenomenon similar to the intervention effects observed with learning diaries [22]. In terms of internal validity, while the randomized controlled design is a key strength, the differential attrition rates between the two groups could potentially bias the results. Although we consider other threats to be minimal, this difference warrants caution in interpretation. Future studies could employ strategies to further equalize participant retention across conditions.

6 Conclusion

This study demonstrates how a process-oriented approach using sequence analysis provides a granular lens to reveal the complex and multifaceted effects of educational interventions on student learning processes. Our findings show that a simple late-of-schedule reminder intervention produced both intended and unintended consequences that would have remained hidden using traditional outcome-based evaluation approaches. The intervention successfully promoted self-regulatory learning behaviors: not only students' time management changed (effect), but students also made strategic trade-offs in the allocation of their learning efforts (side-effect). Simultaneously, our analysis used these methods to pinpoint

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how students without such prompts defaulted to less effective, repetitive content consumption strategies when falling behind. The behavioral divergence between groups underscores a critical insight: interventions targeting scheduling do not merely affect timing but thoroughly reshape how students approach their learning, triggering metacognitive processes that transform surface-level task completion into deeper, more reflective learning engagement. A key component for this effect might be to prompt students to define their own individual deadlines for the course. This way, planning behavior was already scaffolded for all students when the intervention took place, allowing it to trigger the beneficial SRL strategies. This study advocates for the use of methods like sequence analysis not merely as analytical tools, but as exploratory instruments capable of uncovering the hidden curriculum of student behavior and the full, often unexpected, impact of educational interventions.

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